

CS 6371 - Assignment 6

Christian Parker
CXP220034@utdallas.edu

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1. (a)

$$\begin{aligned}
 e &= (\text{or } (\text{not true}) \text{ false}) \\
 &= ((\lambda a. \lambda b. (a ? \text{true} : b)) (\text{not true}) \text{ false}) \\
 &= ((\lambda a. \lambda b. ((a)(\text{true})(b))) (\text{not true}) \text{ false}) \\
 &= ((\lambda a. \lambda b. ((a)(\text{true})(b))) ((\lambda b. (b ? \text{false} : \text{true})) \text{true}) \text{false}) \\
 &= ((\lambda a. \lambda b. ((a)(\text{true})(b))) ((\lambda b. ((b)(\text{false})(\text{true}))) \text{true}) \text{false}) \\
 &= ((\lambda a. \lambda b. ((a)(\lambda x. \lambda y. x)(b))) ((\lambda b. ((b)(\text{false})(\lambda x. \lambda y. x))) (\lambda x. \lambda y. x)) \text{false}) \\
 &= ((\lambda a. \lambda b. ((a)(\lambda x. \lambda y. x)(b))) ((\lambda b. ((b)(\lambda x. \lambda y. y)(\lambda x. \lambda y. x))) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. y))
 \end{aligned} \tag{1}$$

(b)

$$\begin{aligned}
 &((\lambda a. \lambda b. ((a)(\lambda x. \lambda y. x)(b))) ((\lambda b. ((b)(\lambda x. \lambda y. y)(\lambda x. \lambda y. x))) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &((\lambda b. (((\lambda b. ((b)(\lambda x. \lambda y. y)(\lambda x. \lambda y. x))) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. x)(b))) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &(((\lambda b. ((b)(\lambda x. \lambda y. y)(\lambda x. \lambda y. x))) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. x) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &((((\lambda x. \lambda y. x)(\lambda x. \lambda y. y)(\lambda x. \lambda y. x)) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &((((\lambda y. (\lambda x. \lambda y. y)) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &(((\lambda x. \lambda y. y) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &((\lambda y. y) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &(\lambda x. \lambda y. y)
 \end{aligned} \tag{2}$$

2. (a)

$$isempty = \lambda \ell. \text{not } (\pi_1 \ell) \tag{3}$$

(b)

$$\begin{aligned}
 head &= \lambda \ell. \pi_1 (\pi_2 \ell) \\
 tail &= \lambda \ell. \pi_2 (\pi_2 \ell)
 \end{aligned} \tag{4}$$

(c)

$$cons = \lambda e. \lambda \ell. (\text{pair } \text{true } (\text{pair } e \ell)) \tag{5}$$

(d)

$$length = Y(\lambda f. \lambda \ell. (isempty \ell ? 0_{\mathbb{N}} : succ_{\mathbb{N}}(f (tail \ell)))) \tag{6}$$

(e)

$$\begin{aligned}
rev &= Y(\lambda f.(\lambda a.\lambda \ell.(\text{isempty } \ell ? a : f ((\text{head } \ell) :: a) (\text{tail } \ell)))) (\Box) \\
append_helper &= Y(\lambda f.(\lambda \ell_2.\lambda \ell_1.(\text{isempty } \ell_1 ? \ell_2 : f ((\text{head } \ell_1) :: \ell_2) (\text{tail } \ell_1)))) \\
append &= \lambda \ell_1.\lambda \ell_2.(append_helper \ell_2 (rev \ell_1))
\end{aligned} \tag{7}$$

(f)

$$\begin{aligned}
map_helper &= Y(\lambda m.\lambda a.\lambda f.\lambda \ell.(\text{isempty } \ell ? a : m ((f (\text{head } \ell)) :: a) f (\text{tail } \ell))) (\Box) \\
map &= \lambda f.\lambda \ell. rev (map_helper f \ell)
\end{aligned} \tag{8}$$

(g)

$$fold_left = Y(\lambda g.\lambda f.\lambda a.\lambda \ell.(\text{isempty } \ell ? a : g (f) (f a (\text{head } \ell)) (\text{tail } \ell))) \tag{9}$$

3. **Theorem.** If $e \Downarrow e''$ and $e \rightarrow_1 e'$ then $e' \Downarrow e''$.

Proof. There exists a derivation D_L of $e \Downarrow e''$ and a derivation D_S of $e \rightarrow_1 e'$.

We will prove $e' \Downarrow e''$ by structural induction on D_S .

Inductive Hypothesis. If $e_0 \Downarrow e_0''$ has a derivation smaller than D_L and $e_0 \rightarrow_1 e_0'$ has a derivation smaller than D_S , then $e_0' \Downarrow e_0''$.

Inductive Cases.

Case 1. Suppose D_S ends in S1. Then $e = (\lambda v.e_1)e_2$ and $e' = e_1[e_2/v]$.

We must show $e_1[e_2/v] \Downarrow e''$.

Since $e = (\lambda v.e_1)e_2$, D_L must have the following form:

$$D_L = \frac{\lambda v.e_1 \Downarrow \lambda v.e_1 \quad e_1[e_2/v] \Downarrow e''}{(\lambda v.e_1)e_2 \Downarrow e''} \tag{10}$$

$e_1[e_2/v] \Downarrow e''$ is shown by D_L .

Case 2. Suppose D_S ends in S2. Then D_S has the form:

$$D_S = \frac{\frac{D_0}{e_1 \rightarrow_1 e_1'}}{e_1 e_2 \rightarrow_1 e_1' e_2} \tag{11}$$

We must show $e_1' e_2 \Downarrow e''$. Since $e = e_1 e_2$, D_L must have the following form:

$$D_L = \frac{\frac{D_1}{e_1 \Downarrow \lambda v.e} \quad \frac{D_2}{e[e_2/v] \Downarrow e''}}{e_1 e_2 \Downarrow e''} \tag{12}$$

Observe that $e'_1 e_2 \Downarrow e''$ has derivation:

$$\frac{\frac{D'_1}{e'_1 \Downarrow \lambda v.e} \quad \frac{D_2}{e[e_2/v] \Downarrow e''}}{e'_1 e_2 \Downarrow e''} \quad (13)$$

We apply the inductive hypothesis with $D_0 < D_S$, $D_1 < D_L$, $e_0 = e_1$, $e'_0 = e'_1$, and $e''_0 = \lambda v.e$. Therefore, $e'_1 \Downarrow \lambda v.e$, which is D'_1 . We conclude $e'_1 e_2 \Downarrow e''$ from its derivation. $\square$